



PRESEnTATION

TASK:2 API

INTEGRATION & json

HANDLING using

PYTHON

OBJECTIVES:

THE OBJECTIVE OF THIS PROJECT IS TO LEARN HOW PYTHON COMMUNICATES WITH AN EXTERNAL API, RECEIVES DATA IN JSON FORMAT, AND DISPLAYS THE REQUIRED INFORMATION. THIS PROJECT ALSO DEMONSTRATES BASIC ERROR HANDLING WHILE WORKING WITH APIS.

TOOLS USED:

PYTHON 3

PYDROID 3

REQUESTS LIBRARY

RANDOM USER API

```

import requests

# API URL
url = "https://randomuser.me/api/"

try:
    # Fetch data from API
    response = requests.get(url)

    # Check if request was successful
    if response.status_code == 200:

        # Convert JSON data into Python dictionary
        data = response.json()

        # Get user details
        user = data["results"][0]

        print("---- Random User Details ----")
        print("Name :", user["name"]["first"],
              user["name"]["last"])
        print("Gender :", user["gender"])
        print("Email :", user["email"])
        print("Country :", user["location"]["country"])

    else:
        print("Failed to fetch data.")

except:
    print("Check your internet connection or API.")

```

```

IMPORT REQUESTS

TRY:
    URL = "HTTPS://RANDOMUSER.ME/API/"
    RESPONSE = REQUESTS.GET(URL)

    IF RESPONSE.STATUS_CODE == 200:
        DATA = RESPONSE.JSON()

        USER = DATA["RESULTS"][0]

        PRINT("----- RANDOM USER DETAILS -----")
        PRINT("NAME :", USER["NAME"]["FIRST"], USER["NAME"]["LAST"])
        PRINT("GENDER :", USER["GENDER"])
        PRINT("EMAIL :", USER["EMAIL"])
        PRINT("COUNTRY :", USER["LOCATION"]["COUNTRY"])
        ELSE:
            PRINT("FAILED TO FETCH DATA")

    EXCEPT EXCEPTION AS E:
        PRINT("ERROR:", E)

```

EXPLANATION:

THIS PROGRAM USES THE REQUESTS LIBRARY TO SEND A REQUEST TO THE RANDOM USER API. THE API RETURNS USER INFORMATION IN JSON FORMAT. THE PROGRAM CONVERTS THE JSON DATA INTO A PYTHON DICTIONARY AND EXTRACTS THE USER'S NAME, GENDER, EMAIL, AND COUNTRY. IT ALSO CHECKS WHETHER THE API REQUEST WAS SUCCESSFUL AND HANDLES ERRORS USING A TRY-EXCEPT BLOCK.

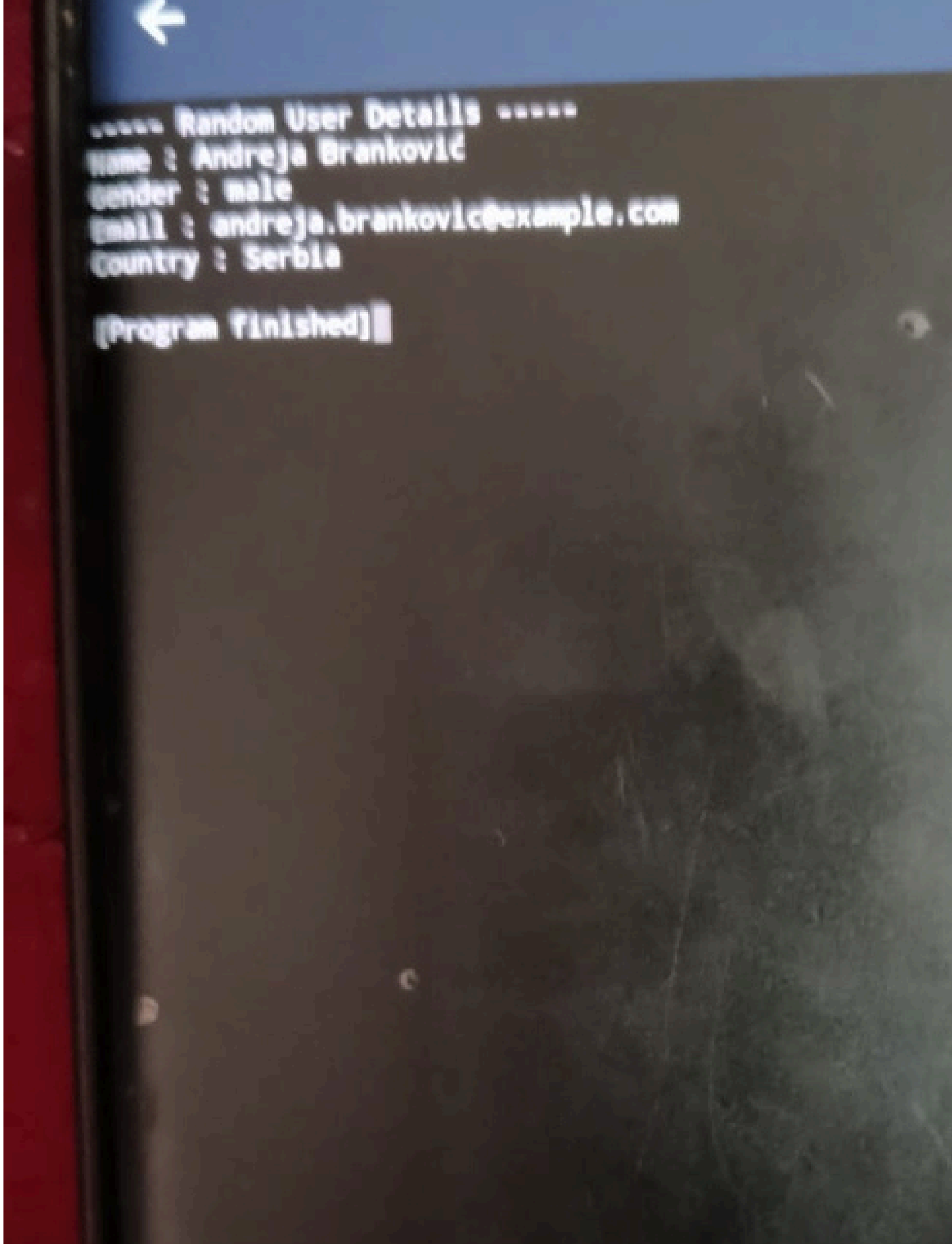
FEATURES:

FETCHES LIVE DATA FROM AN API.

PARSES JSON DATA.

DISPLAYS SELECTED USER INFORMATION.

INCLUDES BASIC ERROR HANDLING.

A screenshot of a terminal window with a dark background. At the top left, there is a white arrow pointing left. The text in the terminal is white and shows the output of a program. It starts with a header 'Random User Details' flanked by asterisks. Below this, it lists four attributes: 'Name : Andreja Brankovic', 'Gender : male', 'Email : andreja.brankovic@example.com', and 'Country : Serbia'. At the bottom, it says '[Program finished]' followed by a cursor.

```
Random User Details *****
Name : Andreja Brankovic
Gender : male
Email : andreja.brankovic@example.com
Country : Serbia

[Program finished]
```

code

output:

Applications:

User profile generation.
Testing web applications.
Learning REST APIs.
JSON data processing.

Conclusion:

This project helped me understand how to integrate an API in Python and work with JSON data. I also learned how to handle errors while fetching online data. API integration is an important skill for developing modern Python applications.

Thank
you

